MB 727 BUSINESS INTELLIGENCE TOOLS
Assignment-2
15 Marks

Note: Answer the follow assignment based on outputs from "R-Software" only. Any claims for mismatch in answers due to results from other tools will not be entertained.

Using the "AirPassengers" data available in "datasets" package of R, answer the following questions (question numbers 1 to 7)

Hint: The AirPassengers data is a time series data. To answer the questions, you need to convert it into a data frame with two columns. The first column should contain the date in "YYYY-MM-DD" format and the second column should contain the count of passengers renamed as "CountPassengers"

1. Which of the following is the correct histogram for the data? **[MSQ, 1 Mark]** *(options will be displayed in Moodle)*

2. Which date has the highest passenger count? *(Note: Enter your answer in YYYY-MM-DD format)* **[1 Mark]**

3. How many data points have a passenger count greater than or equal to 200 and less than or equal to 500? **[1 Mark]**

4. What is the mean value for the data set generated in Question-3? *(Note: Enter your answer rounded to two decimal places. For example, if your answer is "123.456" enter it as "123.46")* **[1 Mark]**

5. You are asked to perform a statistical test to see if the mean value in Question-4 **is SIGNIFICANTLY HIGHER (at an alpha of 15%)** than the entire data's mean (assume the entire data is the population). Then, what is the value of the computed test statistic? *(Note: Enter your answer rounded to two decimal places. For example, if your answer is "123.456" enter it as "123.46")* **[1 Mark]**

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6. You are asked to perform a statistical test to see if the mean value in Question-4 **is SIGNIFICANTLY HIGHER (at an alpha of 15%)** than the entire data's mean (assume the entire data is the population). Then, what is the p-value for test? *(Note: Enter your answer rounded to five decimal places. For example, if your answer is "123.456789" enter it as "123.45679")* **[1 Mark]**

7. You are asked to perform a statistical test to see if the mean value in Question-4 **is SIGNIFICANTLY HIGHER (at an alpha of 15%)** than the entire data's mean (assume the entire data is the population). Then, what do you conclude? **[1 Mark]**

   a) Reject the Null
   b) Do not reject the Null

Using the "Titanic" data available in "datasets" package of R, answer the following questions (question number 8 to 11)

Hint: The Titanic data is an array data. To answer the questions, you need to convert it into a data frame with four columns.

8. How many children on board the titanic survived? **[1 Mark]**

9. In the 1997 "TITANIC" movie directed by James Cameron, it was suggested that the women and children were given preference for offboarded before the men. If this were true, the proportion of men who survived should be lesser than the proportion of women and children who survived. I am tasking you (a data science aspirant) to statistically verify this claim. Then, what is the value of the computed test statistic (for an 80% Confidence interval)?

   (*Hint: This is not a straight forward question. You need to understand what type of test is to be performed, and then appropriately generate the result. Your test must consider "with continuity correction")*

   (Note: Enter your answer rounded to two decimal places. For example, if your answer is "123.456" enter it as "123.46") **[2 Marks]**

10. What is the upper bound of the 80% confidence interval created for question 11? *(Note: Enter your answer rounded to four decimal places. For example, if your answer is "123.45678" enter it as "123.4568")* **[1 Mark]**

11. In the 1997 "TITANIC" movie directed by James Cameron, it was suggested that the women and children were given preference for offboarded before the men. If this were true, the proportion of men who survived should be lesser than the proportion of women and children who survived. I am tasking you (a data science aspirant) to statistically

verify this claim. Then, what is conclusion of the test (for an 80% Confidence interval)? **[Select all that is applicable, 1 Mark]**

a. Reject the Null and claim that the proportion of men who survived is not significantly different (lower) from the proportion of women and children who survived

b. Do not reject the Null and claim that the proportion of men who survived is not significantly different (lower) from the proportion of women and children who survived

c. Reject the Null and claim that the proportion of men who survived is significantly different (lower) from the proportion of women and children who survived

d. Do not reject the Null and claim that the proportion of men who survived is significantly different (lower) from the proportion of women and children who survived

It is disheartening to know that (some) students of NIT-T are indulging in shoplifting activities at the 2K super market. Intelligence has no value when moral and ethical values are absent in students. The data in the file "A2_Data_Q12_Q17.csv" provides a hypothetical information of past sales and theft that have happened at some other supermarket (not 2K market). Give this data, answer the following questions (question numbers 12 to 17)

12. A simple linear regression model is fit to the data set with an aim of predicting the possible theft for a given sales. Then, what is the intercept value? *(Note: Enter your answer rounded to four decimal places. For example, if your answer is "123.45678" enter it as "123.4568")* **[0.5 Mark]**

13. A simple linear regression model is fit to the data set with an aim of predicting the possible theft for a given sales. Then, what is the slope of the model? *(Note: Enter your answer rounded to four decimal places. For example, if your answer is "123.45678" enter it as "123.4568")* **[0.5 Mark]**

14. A simple linear regression model is fit to the data set with an aim of predicting the possible theft for a given sales. Then, what is the magnitude and direction of correlation between "Sales" and "Theft"? *(Note: Enter your answer rounded to four decimal places. For example, if your answer is "123.45678" enter it as "123.4568")* **[0.5 Mark]**

15. A simple linear regression model is fit to the data set with an aim of predicting the possible theft for a given sales. Then, what is the p-value for the model? *(Note: Enter your answer rounded to four decimal places. For example, if your answer is "123.45678" enter it as "123.4568")* **[0.5 Mark]**

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16. A simple linear regression model is fit to the data set with an aim of predicting the possible theft for a given sales. Then, what is the total variability in "Theft" that is explained by "Sales"? *(Note: Enter your answer in Percentage rounded to two decimal places without the "%" symbol. For example, if your answer is "12.345%" enter it as "12.35")* **[0.5 Mark]**

17. A simple linear regression model is fit to the data set with an aim of predicting the possible theft for a given sales. Then, at a 95% confidence level, what do you conclude? **[Select all that is applicable, 0.5 Mark]**

    a. Reject the Null and conclude that sales can explain Theft significantly

    b. Do Not reject the Null and conclude that sales can explain Theft significantly

    c. Reject the Null and conclude that sales cannot explain Theft significantly

    d. Do Not reject the Null and conclude that sales cannot explain Theft significantly

- - THE END - -